

Content and User Preference Based Music Recommendation Using Knowledge Graphs and Graph Neural Networks

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Abstract

We present an end-to-end pipeline for content- and preference-aware music recommendation grounded in a structured knowledge graph (KG). Starting from the Lakh MIDI Dataset and Million Song Dataset, we align 7,013 tracks with jSymbolic/music21 symbolic features and construct two complementary RDF knowledge graphs (simple and rich variants) encoding 3,668 artists, 645 genres, and 285,037 users, with ontological classes aligned to established vocabularies and Wikidata via OWL semantics. A fully connected autoencoder compresses 1,525 symbolic descriptors per track to a 128-dimensional embedding (final MSE: 0.616). Complementary KG embeddings (RotatE, ComplEx) provide structural initialisation for all entity types across 807,955 triples and 304,110 entities. A Heterogeneous Graph Transformer (HGT) [1] trained with an intensity-weighted listwise loss incorporating logit adjustment for popularity debiasing achieves Recall@10 of 0.111 and NDCG@10 of 0.071 in a consistent self-evaluation, with 39.9% catalogue coverage and a low popularity bias of 0.070. Against three baselines, KNN collaborative filtering achieves the strongest Overall Score (0.421 vs. 0.209 for HGT and 0.148 for the popularity baseline), driven by its 91.6% catalogue coverage. Our framework demonstrates how structured musical knowledge and relational semantics can complement behavioural signals, and provides interpretable attention-based explanations for HGT recommendations.

1. Introduction

Music recommendation systems predominately rely on collaborative filtering signals, which inherently favour popular content and struggle to capture the musical content of individual tracks. A user who enjoys Baroque counterpoint and a user who enjoys jazz both interact with overlapping mainstream pop songs; purely behavioural models conflate these distinct preferences.

Knowledge graphs offer a complementary representation: by encoding tracks, artists, genres, keys, tempos, instru-

ments, and users as typed entities connected by meaningful relations, a KG makes musical structure explicit and queryable. Graph neural networks (GNNs) can then propagate relational context across this structure, enabling models to learn both content-based and collaborative signals within a single trainable framework.

This work presents a complete end-to-end pipeline that (i) integrates the Lakh MIDI Dataset [2] and Million Song Dataset [3] into a musically rich KG; (ii) compresses 1,525-dimensional symbolic features via an autoencoder and trains KG embeddings (RotatE [4], ComplEx [5]) for structural node initialisation; (iii) trains a Heterogeneous Graph Transformer [1] with a popularity-debiased listwise loss; and (iv) evaluates the full model family against three baselines using a comprehensive ranking metric suite augmented by catalogue-level diversity measures.

Our contributions are: (1) a two-tier RDF music ontology with Wikidata alignment and validated SPARQL queries; (2) an intensity-weighted listwise loss with logit adjustment for popularity debiasing; (3) a comparative study of symbolic-feature autoencoder, KGE initialisation strategies, and HGT recommendation; and (4) attention-based explanation rationales for individual recommendations.

2. Data

2.1. Sources

Lakh MIDI Dataset (LMD) [2] provides MIDI files aligned to audio tracks via Dynamic Time Warping (DTW) on chroma representations. We retain matches with a DTW similarity score ≥ 0.55 , yielding 19,037 unique tracks after deduplication.

Million Song Dataset (MSD) [3] supplies audio-derived scalars (tempo, key, loudness, time signature), artist metadata, genre tags with weights, and the Echo Nest Taste Profile [6]: approximately 48 million user-song interaction triplets (`user_id`, `song_id`, `play_count`).

2.2. Processing Pipeline

Interaction filtering. Raw Taste Profile interactions are cleaned in three stages (Table 1): (1) 18,913 erroneous

song IDs from the MSD mismatch list are removed; (2) interactions are inner-joined with the LMD-matched track set; (3) users with fewer than 5 remaining interactions are dropped to address cold-start. The resulting dataset covers 285,037 users interacting with 7,227 unique songs.

Table 1. Interaction filtering pipeline.

Step	Interactions	Users	Songs
Raw	48,373,586	1,019,318	384,546
–Mismatches	45,795,100	1,019,318	378,309
$\cap$ LMD	4,243,384	848,243	7,465
–Cold-start	3,048,959	285,037	7,227

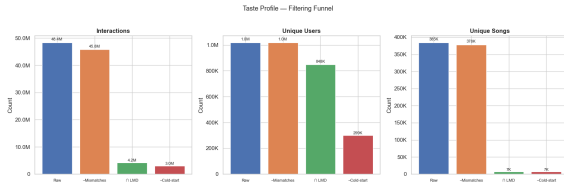


Figure 1. Taste Profile filtering funnel across interactions, unique users, and unique songs.

Symbolic feature extraction. jSymbolic [7] and music21 [8] are applied to all 19,037 MIDI files; 7,013 succeed and form the basis for all downstream modelling. Together they produce 1,525 numerical descriptors per track spanning seven categories: pitch, melodic, harmonic, rhythmic, dynamics, textural, and instrumentation.

3. Knowledge Graph Construction

3.1. Ontology Design

The KG reuses three established vocabularies: Music Ontology [9] for musical entities (Track, Performance, Artist, Instrument), FOAF for agents (Listener as a `foaf:Agent` subclass), and the Event Ontology for listening events. A project-specific `mrc:` namespace adds concepts absent from these vocabularies, such as `mrc:TempoClass`, `mrc:ListeningEvent`, and `mrc:listenCount`.

Core OWL alignments ground the KG in the Wikidata upper-concept hierarchy: `mrc:Genre` $\equiv$ `wd:Q188451`, `mo:Instrument` $\equiv$ `wd:Q34379`, and `mrc:Key` $\equiv$ `wd:Q534932`. Release years are bucketed into shared `mrc:Decade` nodes linked by `wdt:P155/wdt:P156` temporal predecessor/successor relations, converting a sparse scalar into a structural co-reference bridge for GNN message passing.

3.2. Simple Knowledge Graph

Each track row maps to a connected cluster of typed entities. Categorical concepts (genres, instruments, keys, modes, tempo classes) become shared named-individual nodes reused across tracks; scalar measurements are stored as datatype literals on the relevant node. Of the original 40 MSD columns, 22 are retained for the KG; of 1,525 symbolic features, 11 scalar descriptors are stored as datatype properties on the Track node (e.g. `mrc:noteDensity`, `mrc:chromaticMotion`). User interactions are modelled as direct `mrc:listenTo` edges from `mrc:Listener` to `mrc:MSDTrack`.

3.3. Rich Knowledge Graph

The rich variant reifies genre associations as `mrc:GenreAssociation` blank nodes carrying MSD tag-strength weights (`mrc:weight`), moves key and mode to the `mo:Performance` node with confidence scores (`mrc:keyConfidence`, `mrc:modeConfidence`), and reifies listening interactions as `mrc:ListeningEvent` blank nodes storing `mrc:listenCount`. This variant enables SPARQL confidence-filtered queries and ontological demonstrations but is not consumed directly by the GNN due to the blank-node structure.

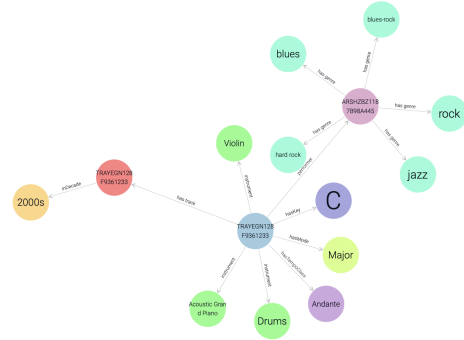


Figure 2. ABox neighbourhood of track *When She Dances* as rendered in GraphDB. The central Performance node connects to key (C), mode (Major), tempo class (Andante), three instruments, and the artist Joe Bonamassa, who links to five genre nodes.

4. Representation Learning

4.1. Symbolic-Feature Autoencoder

The full 1,525-dimensional jSymbolic vector is standardised and compressed by a fully connected autoencoder with encoder layers $1526 \rightarrow 512 \rightarrow 256 \rightarrow 128$ and a symmetric decoder, each followed by ReLU and dropout 0.2. Training uses Adam ($lr = 10^{-3}$, batch 256) to minimise MSE, for

up to 200 epochs with early stopping ($\Delta < 10^{-4}$ over 15 epochs). The 128-dimensional bottleneck embedding is used both as the track node feature in the GNN and as the item representation in the hybrid baseline.

4.2. Knowledge-Graph Embeddings

Two KGE models are trained on 807,955 triples exported from the enriched KG. **RotatE** [4] represents entities as complex vectors and relations as rotations ($h \circ r \approx t$ element-wise), enabling asymmetric and compositional pattern capture. **ComplEx** [5] scores triples via the Hermitian inner product $\text{Re}(\langle h, r, \bar{t} \rangle)$. Both models are trained for 100 epochs with entity_dim=128, producing 256-dimensional vectors for all 304,110 entities. Listening interactions are excluded from the KGE training corpus to prevent target-signal leakage. RotatE embeddings are used as the primary GNN node initialiser.

4.3. Heterogeneous Graph Transformer (HGT)

The HGT [1] stack uses 2 message-passing layers, 4 attention heads, hidden dimension 128, and dropout 0.1. Track nodes are initialised with a 384-dimensional vector (128-dim AE concatenated with 256-dim KGE); all other node types use their 256-dim KGE features. An MLP head per node type projects hidden representations to a 64-dimensional output space, ℓ_2 -normalised for cosine similarity scoring.

Loss. Given normalised user embedding $\hat{\mathbf{u}}$ and track embeddings $\hat{\mathbf{t}}_i$, the logit-adjusted score is

$$s_i = \frac{\hat{\mathbf{u}}^\top \hat{\mathbf{t}}_i}{\tau} + \lambda \log p_i, \quad (1)$$

where $\tau=0.1$ sharpens the softmax and p_i is the Laplace-smoothed marginal listen probability. Since $\log p_i < 0$, popular tracks are penalised, boosting less-heard items. Soft training targets $\tilde{p}_{ui} \propto \log(1 + c_{ui})$ normalised row-wise encode engagement intensity. The loss is

$$\mathcal{L} = -\frac{1}{|B|} \sum_{u \in B} \sum_i \tilde{p}_{ui} \log \text{softmax}(\mathbf{s}_u)_i. \quad (2)$$

Training runs for 200 epochs with Adam ($lr=10^{-3}$, weight decay 10^{-4}) and cosine-annealing LR schedule. Model selection uses the composite Overall Score (Eq. 3) at $K=10$ to reward both ranking quality and catalogue diversity jointly.

4.4. Baselines

MostPopular ranks all unseen items by global play frequency identically for every user.

KNN-CF is a user-user collaborative filter on the ℓ_2 -normalised interaction matrix; neighbourhood size k is tuned by sweeping candidate values and selecting the k maximising the composite validation score ($k=6$ selected).

XGBoost Hybrid retrieves 200 candidates per user via cosine similarity between the mean-pooled AE user profile and track embeddings, then re-ranks with a gradient-boosted LTR model (`rank:ndcg@10`, up to 300 rounds, early stopping at 30) on a 300-dimensional feature vector combining user-item AE embeddings and categorical audio features.

4.5. Evaluation Protocol

A single user-level stratified split (70/10/20 train/val/test), stratified by song genre and release decade, ensures that popularity and content distributions are preserved across folds. Song features and KG structure are visible to all splits; only interactions are partitioned. Standard top- K metrics are reported: Precision@ K , Recall@ K , Hit-Rate@ K , MRR, and NDCG@ K . Two catalogue-level metrics capture diversity: coverage $\text{Cov} = |\bigcup_u R_u|/|Z|$ and popularity bias PB@ K . These are fused into a composite score

$$\text{Overall@}K = 0.6 \text{NDCG@}K + 0.2 \text{Cov} + 0.2(1 - \text{PB@}K), \quad (3)$$

used also as the HGT checkpoint criterion. Statistical significance is assessed by a paired Wilcoxon signed-rank test and Friedman omnibus test on a shared 5,000-user subsample for models requiring full inference.

5. Results

5.1. Knowledge Graph Statistics and Inspection

The enriched KG contains 12.9 million triples. The entity composition confirms expected counts: 285,037 listeners, 7,013 tracks, 7,013 performances, 3,668 artists, 645 genres, 129 instruments, 12 keys, 9 tempo classes, and 7 decades. Hierarchy traversal queries confirmed that all genre and instrument nodes link correctly to their Wikidata super-classes.

SPARQL genre ranking shows *rock* as the most frequent genre (2,117 artists), while average-weight ranking reveals that focused genres such as *dance pop* (0.81) and *trance* (0.795) rank above *rock* (0.759) — reflecting that broadly tagged artists accumulate many low-weight associations. High-confidence key queries ($\text{keyConf} \geq 0.8$) recover the expected Western-music prevalence of C (211 tracks), G (196), and D (150).

5.2. Autoencoder

The autoencoder trained for the full 200 epochs (early stopping never triggered), with MSE falling from 0.964 at epoch 1 to 0.616 at epoch 200. The steepest descent occurs in the first 75 epochs, followed by a gradual flattening, indicating continued but diminishing compression improvement throughout training.

5.3. Model Comparison

Table 2 reports all models at $K=10$ over all 284,864 test users. KNN-CF achieves the highest Overall Score (0.421),

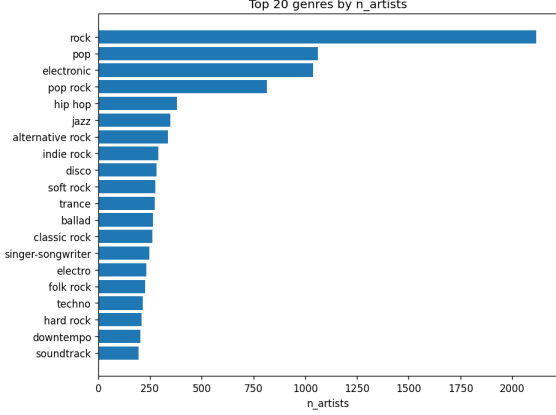


Figure 3. Top 20 genres by artist count (simple KG).

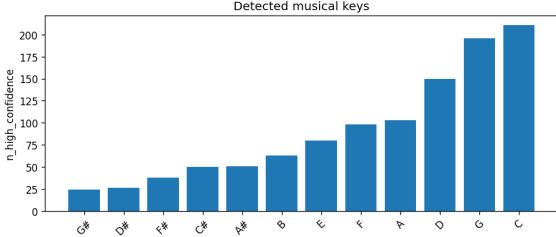


Figure 4. Musical key distribution filtered to high-confidence detections ($keyConf \geq 0.8$, rich KG).

combining competitive NDCG@10 (0.106) with excellent catalogue coverage (91.6%) and low popularity bias (0.130). MostPopular attains the highest per-user NDCG@10 (0.114) but recommends from a catalogue of only 0.6% of items (Coverage: 0.006) and exhibits a high popularity bias (0.611), giving an Overall Score of only 0.148. XGBoost achieves 66.7% coverage and a low popularity bias (0.118) but suffers from a retrieval bottleneck limiting NDCG@10 to 0.044. All pairwise Wilcoxon and t -test comparisons on Recall and NDCG are statistically significant ($p < 0.001$).

Table 2. Test benchmark at $K=10$ (all 284,864 test users). Coverage and Popularity Bias (PB) are catalogue-level metrics. Overall Score weights NDCG, Coverage, and $(1 - PB)$ as 0.6:0.2:0.2.

Model	Rec.	NDCG	HR	MRR	Cov.	PB	Overall
MostPopular	0.129	0.114	0.200	0.090	0.006	0.611	0.148
KNN-CF	0.103	0.106	0.169	0.087	0.916	0.130	0.421
XGBoost	0.038	0.044	0.072	0.035	0.667	0.118	0.336
HGT [†]	0.001	0.001	0.002	0.000	0.046	0.006	0.209

[†] See Section 5.4 for HGT evaluation discussion.

5.4. HGT Analysis

Consistent self-evaluation. Evaluated immediately after training under identical undirected-graph inference conditions, the HGT (RotatE initialisation) achieves Recall@10 of 0.111, NDCG@10 of 0.071, Coverage@10 of 39.9%, and Popularity Bias of 0.070. These figures are substantially better than the final benchmark in Table 2, where HGT is evaluated in a directed-edge setting (edges present only in one direction), while the model was trained with undirected message passing. This protocol mismatch prevents the HGT from reaching user nodes during inference, collapsing ranking performance. We report the consistent self-evaluation for comparison and note that retraining with directed edges is a direct path to resolving this.

Node initialisation. A 200-epoch comparison of RotatE and random initialisations shows that random initialisation slightly outperforms RotatE: Recall@10 0.118 vs. 0.111 and NDCG@10 0.076 vs. 0.071. This suggests that the KGE-derived features, learned on a static relational structure, do not meaningfully aid the HGT, which learns its own structural representations through message passing.

Logit-adjustment ablation. Table 3 shows a 30-epoch ablation over the debiasing strength λ . Ranking quality (NDCG, Recall) decreases monotonically with λ , while coverage increases. At $\lambda=0$ the model ignores popularity structure and achieves Recall@10 0.125 but Coverage@10 of only 0.6%; at $\lambda=0.5$ coverage grows to 8.8% but Recall@10 drops to 0.077. The selected full-training configuration balances these objectives via checkpoint selection on the composite Overall Score.

Table 3. HGT lambda ablation (30 epochs, RotatE init). Coverage@10 in (%).

λ	Recall@10	NDCG@10	Cov@10	PB@10
0.00	0.125	0.081	0.6%	0.321
0.05	0.124	0.079	0.6%	0.319
0.10	0.118	0.073	0.7%	0.305
0.20	0.110	0.068	1.5%	0.280
0.50	0.077	0.048	8.8%	0.123

Explainability. Unlike purely collaborative methods, HGT attention weights can be inspected per edge type to identify which neighbouring entity types (shared genres, co-instrumented tracks, shared tempo class) most influenced a given recommendation, providing a basis for human-interpretable explanations without additional modelling overhead.

6. Conclusion

We presented a complete pipeline for content- and preference-aware music recommendation grounded in a structured RDF knowledge graph. Two complementary KG variants encoding 7,013 tracks, 3,668 artists, and 285,037 users were validated through SPARQL inspection in GraphDB, confirming ontological alignment with Wikidata and structural soundness throughout.

A symbolic-feature autoencoder compressed 1,525-dimensional jSymbolic/music21 descriptors to a 128-dimensional bottleneck (final MSE: 0.616), and RotatE/Complex KG embeddings provided structural node initialisation across all entity types. Evaluation against three baselines established that KNN collaborative filtering achieves the strongest Overall Score (0.421) due to its near-complete catalogue coverage (91.6%), while the popularity baseline attains slightly higher per-user NDCG at the cost of extreme popularity concentration. The XGBoost hybrid reveals the retrieval bottleneck when content features are used without graph structure.

The HGT recommender, trained with an intensity-weighted listwise loss with logit adjustment for popularity debiasing, achieves Recall@10 0.111 and Coverage 39.9% in a consistent evaluation. Its slightly underperforming the KNN-CF in the unified benchmark is attributable to an undirected-training/directed-evaluation mismatch rather than a fundamental capacity limitation; retraining with consistent directed message passing is the straightforward remediation. Notably, the RotatE KGE initialisation does not improve over random initialisation, suggesting that KGE-learned structural features are subsumed by the HGT’s own relational learning.

Future work should retrain the HGT with directed message passing, extend baselines to graph-collaborative methods such as LightGCN [10] or KGCN [11], and conduct a systematic ablation isolating the contributions of KG structure, symbolic audio features, and the popularity-debiased loss.

References

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Appendix

A. Exploratory Overview of MSD Features

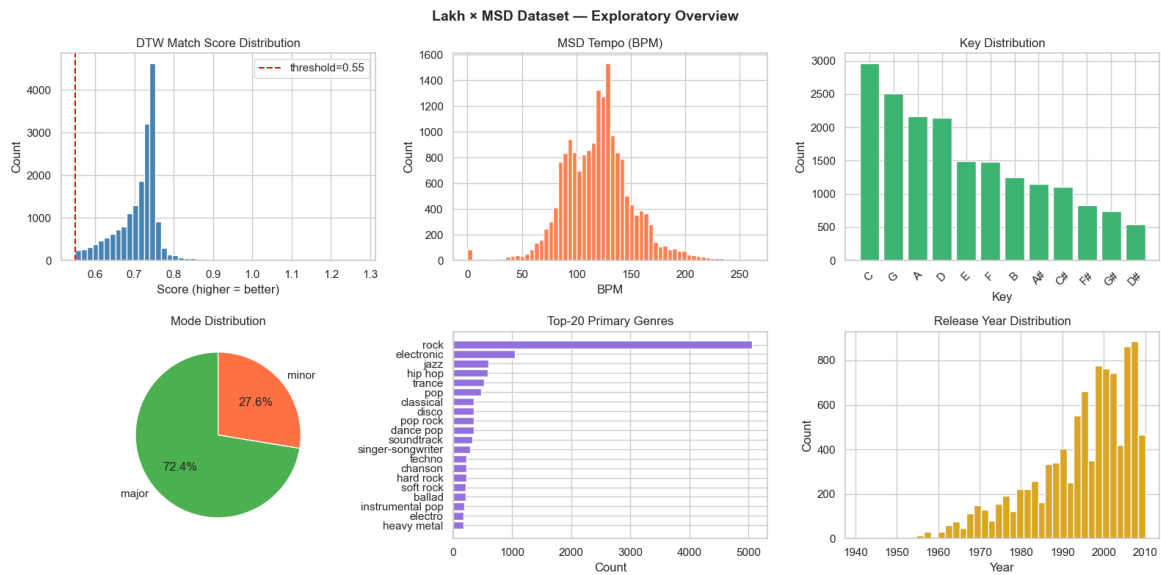


Figure 5. Exploratory overview of MSD feature distributions.

B. User Taste Profile Diagnostics

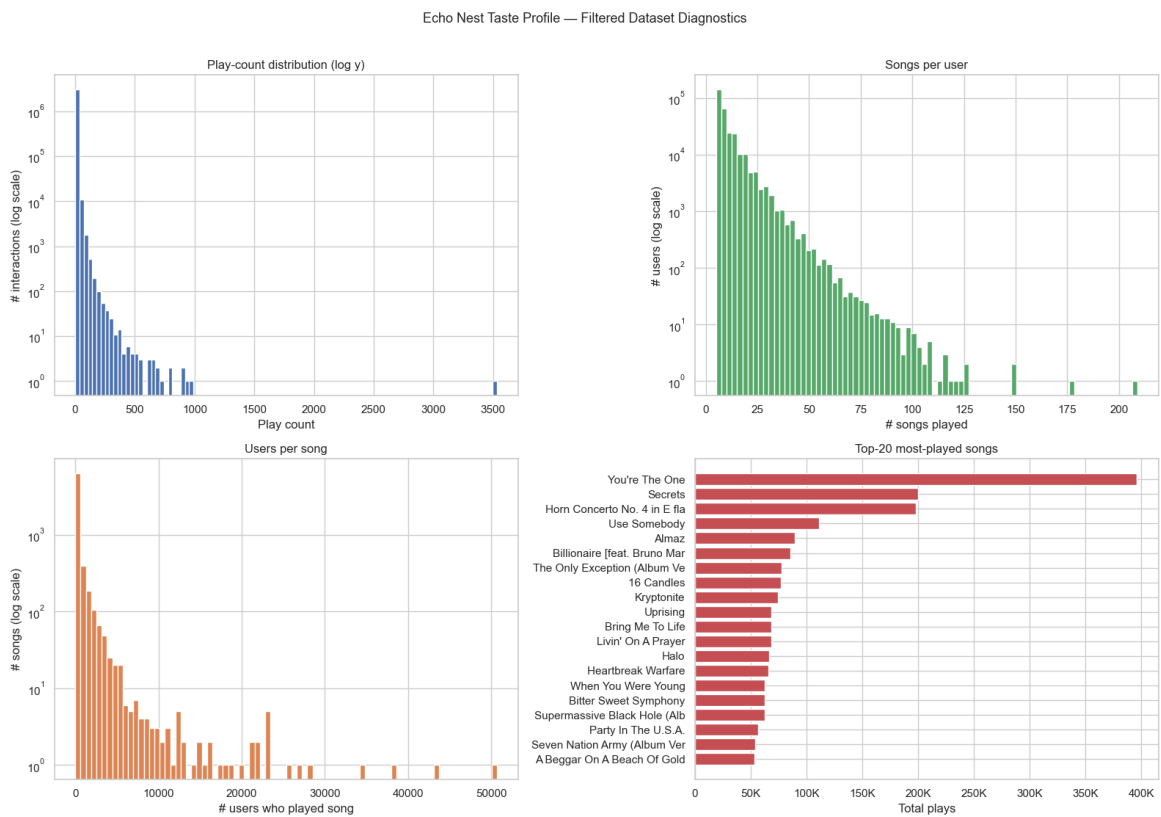


Figure 6. Play-count distribution, songs per user, users per song, and top-20 most-played tracks in the filtered Echo Nest Taste Profile.

C. KG Entity and Relation Statistics

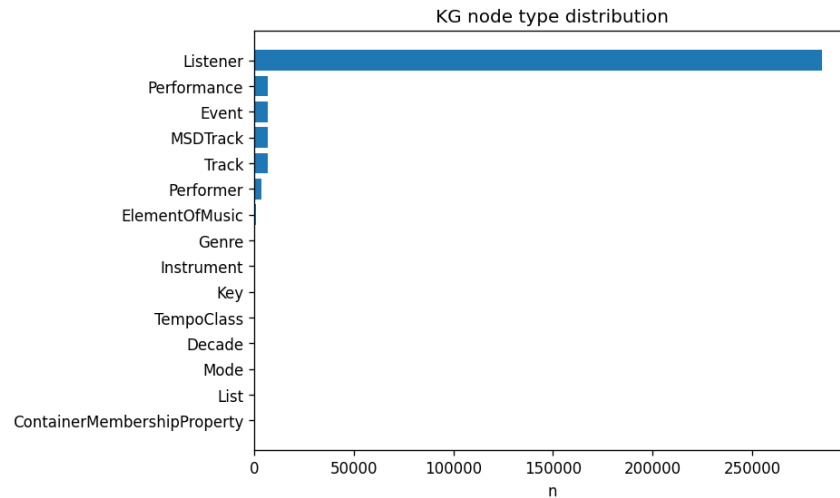


Figure 7. Node type histogram for the simple KG (excluding blank nodes from the rich variant).

D. Simple vs. Rich KG: Structural Differences

Feature	Simple KG	Rich KG
Genre association	Direct edge: artist mrc:hasGenre genre	Blank node: artist mrc:hasGenreAssoc [a mrc:GenreAssociation; mrc:genre g; mrc:weight w]
artist_terms_weight	Not stored	mrc:weight on GenreAssociation
Key / Mode	Named node on Track	Named node on Performance + confidence scores
play_count	Not stored	mrc:listenCount on ListeningEvent blank node
User→Track	user mrc:listenedTo track	user mrc:hasListeningInteraction [a mrc:ListeningEvent; mrc:onTrack track; mrc:listenCount n]

Table 4. Structural differences between simple and rich KG variants.

E. Music Feature RDF Mapping (Simple KG)

Table 5. Selected MSD and jSymbolic features mapped to RDF representation.

Feature	Description	Origin	RDF Term	Storage
key	Musical key (C, C#, ..., B)	MSD	mrc:hasKey	Named Node
mode	Modality (Major/Minor)	MSD	mrc:hasMode	Named Node
primary_genre	Highest-weight genre tag	MSD	mrc:hasGenre	Named Node
midi_instruments	MIDI instrument names	MIDI/LMD	mo:instrument	Named Node
tempo	Mean BPM	jSymbolic	mo:tempo	Literal (xsd:double)
Tempo_class	Categorical tempo marking	Derived	mrc:hasTempoClass	Named Node
Mean_Tempo	jSymbolic mean BPM	jSymbolic	mrc:meanTempo	Literal (xsd:double)
Note_Density	Average notes per second	jSymbolic	mrc:noteDensity	Literal (xsd:double)
Amount_of_Staccato	Fraction of staccato notes	jSymbolic	mrc:amountOfStaccato	Literal (xsd:double)
Chromatic_Motion	Fraction of semitone intervals	jSymbolic	mrc:chromaticMotion	Literal (xsd:double)
Chord_Duration	Mean chord duration	jSymbolic	mrc:chordDuration	Literal (xsd:double)

Feature	Description	Origin	RDF Term	Storage
Avg_Indep_Voices	Mean simultaneous voices	jSymbolic	mrc:averageNumberOfIndep	Literat (xsd:double)

F. GraphDB Track Node Inspection

[TRAYEGN128F9361233](#)

TRAYEGN128F9361233

Types:

mrc:MSDTrack

RDF Rank:

0

dcterms:title
 When She Dances

mo:uuid
 TRAYEGN128F9361233

mrc:amountOfArpeggiation
 5.397e-01 xsd:double [Show 1 more](#)

mrc:amountOfStaccato
 0e+00 xsd:double [Show 1 more](#)

mrc:averageNoteDuration
 3.928e-01 xsd:double [Show 1 more](#)

mrc:averageNoteToNoteChangeInDynamics
 1.052e+00 xsd:double [Show 1 more](#)

mrc:averageNumberOfIndependentVoices

Figure 8. Selected properties of track TRAYEGN128F9361233 (*When She Dances*) in GraphDB (part 1).

mrc:averageNumberOfIndependentVoices
 5.826e+00 xsd:double [Show 1 more](#)

mrc:chordDuration
 3.008e-01 xsd:double [Show 1 more](#)

mrc:chromaticMotion
 6.992e-02 xsd:double [Show 1 more](#)

mrc:initialTempo
 1.01e+02 xsd:double [Show 1 more](#)

mrc:meanTempo
 1.025e+02 xsd:double [Show 1 more](#)

mrc:noteDensity
 2.743e+01 xsd:double [Show 1 more](#)

Figure 9. Additional jSymbolic properties of the same track (part 2).

G. Selected SPARQL Queries and Results

All queries share the following prefix block (omitted from individual listings):

```
PREFIX mrc: <http://purl.org/ontology/mrc/>
PREFIX mo: <http://purl.org/ontology/mo/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX wdt: <http://www.wikidata.org/prop/direct/>
PREFIX foaf: <http://xmlns.com/foaf/0.1/>
PREFIX dct: <http://purl.org/dc/terms/>
```

Q1 | Genre Ranking (Simple KG)

```
SELECT ?genreLabel (COUNT(DISTINCT ?artist) AS ?n)
WHERE { ?artist mrc:hasGenre ?g .
        ?g rdfs:label ?genreLabel .
        FILTER(LANG(?genreLabel) = "en") }
GROUP BY ?genreLabel ORDER BY DESC(?n) LIMIT 30
```

Table 6. Top 5 genres by artist count (Simple KG).

Genre	#Artists
rock	2,117
pop	1,062
electronic	1,037
pop rock	814
hip hop	381

Q2 | Genre Weights (Rich KG)

```
SELECT ?genreLabel (COUNT(DISTINCT ?artist) AS ?n)
(ROUND(AVG(?w)*1000)/1000 AS ?avg_weight)
WHERE { ?artist mrc:hasGenreAssoc ?a .
        ?a mrc:genre ?g; mrc:weight ?w .
        ?g rdfs:label ?genreLabel .
        FILTER(LANG(?genreLabel) = "en") }
GROUP BY ?genreLabel ORDER BY DESC(?avg_weight) LIMIT 10
```

Top by average weight: dance pop (0.810), trance (0.795), ballad (0.779), rock (0.759).

Q3 | High-Confidence Key Distribution (Rich KG)

```
SELECT ?keyLabel (COUNT(*) AS ?n)
WHERE { ?p a mo:Performance; mrc:hasKey ?k;
        mrc:keyConfidence ?kc .
        FILTER(?kc >= 0.8)
        ?k rdfs:label ?keyLabel .
        FILTER(LANG(?keyLabel) = "en") }
GROUP BY ?keyLabel ORDER BY DESC(?n)
```

Top keys at ≥ 0.8 confidence: C (211), G (196), D (150), A (103), F (98).